Week 2 Report: Structuring Data & ETL Process (Full Depth with Validation & Diagrams)

📃Page 1: Understanding & Identifying Data Issues

Step 1: Exploring the Raw Datasets

Before designing the ETL process or creating a master table, it was essential to analyze the six raw datasets to fully understand their structure, column types, relationships, and overall purpose. The goal was to identify which datasets could be integrated, what fields were relevant, and how they connect through foreign keys or shared values.

Datasets Used:

1. **learneropp1**
2. Contains mappings between learners and opportunities.
3. Includes fields: `enrollment_id`, `learner_id`, `assigned_cohort`, `apply_date`, `status`.
4. **learner1**
5. Describes learners' educational background.
6. Fields: `learner_id`, `degree`, `institution`, `major`.
7. **cohort1**
8. Contains details of learning cohorts.
9. Fields: `cohort_code`, `start_date`, `end_date`.
10. **cognito2**
11. Authentication and profile data for learners.
12. Fields: `user_id`, `email`, `gender`, `UserCreateDate`, `UserLastModifiedDate`, `birthdate`, `city`, `zip`, `state`.
13. **opportunity1**
14. Details about opportunities (internships, jobs, etc.).
15. Fields: `opportunity_id`, `opportunity_name`, `category`, `opportunity_code`.

16. *Note:* This table had a mislabeled column (`learner_id`) that actually contained `opportunity_id` values.

17. **marketing1**

18. Digital campaign performance data.

19. Fields: `reach`, `clicks`, `impressions`, `cost`, `campaign_id`, `platform`.
20. *Note:* This dataset was excluded from the master table due to lack of relevant linking keys.

🪧 Relationship Analysis (Before and After Correction)

Originally, several relationships were misinterpreted due to naming mismatches and inconsistent ID structures. After correction, the relationships between tables were finalized as follows:

Corrected Relationships Diagram (Textual):

```
learneropp1.enrollment_id → learner1.learner_id
learner1.learner_id       → cognito2.user_id
learneropp1.assigned_cohort → cohort1.cohort_code
learneropp1.learner_id      → opportunity1.opportunity_id
```

These mappings were implemented using LEFT JOINs in PostgreSQL.

🛠️ Step 2: Identifying Data Quality Issues

To ensure the accuracy of the ETL process, we checked for the following issues using SQL queries:

1. Missing Values:

```sql
SELECT COUNT(*) FROM master_table_raw WHERE email IS NULL;
```

- Emails were missing due to broken joins between `cognito2` and `learner1`.

2. Duplicate Records:

```sql
SELECT enrollment_id, COUNT(*) FROM learneropp1
GROUP BY enrollment_id HAVING COUNT(*) > 1;
```

- Some `enrollment_id` values were repeated.

3. Inconsistent Formats:

- Timestamps in `cohort1` and `cognito2` were in scientific notation or ISO 8601 format, causing errors on import.

4. Orphan Records:

```
SELECT assigned_cohort FROM learneropp1
WHERE assigned_cohort NOT IN (SELECT cohort_code FROM cohort1);
```

- `assigned_cohort` contained values with no matching record in `cohort1`.

📄Step 3: Documenting Findings

Findings were documented to inform the ETL process. Issues like `NULL` values in key columns, orphan keys, and malformed timestamps were all addressed either by filtering or correcting joins.

📆Page 2: Building the Master Table & ETL Process

Step 1: Master Table Planning

The Master Table is intended to serve as a unified dataset for all future analysis and visualization tasks. Its schema combines key learner information, profile details, opportunity data, and cohort timing.

Schema Planning Table:

Field	Source Table	Type	Description
enrollment_id	learneropp1	text	Unique enrollment identifier
learner_id	learneropp1	text	Learner UUID
assigned_cohort	learneropp1	text	Cohort assigned to learner
apply_date	learneropp1	timestamp	Date learner applied
status	learneropp1	text	Current enrollment status
degree	learner1	text	Academic degree
institution	learner1	text	Name of institution
major	learner1	text	Area of study
cohort_code	cohort1	text	Cohort unique code
start_date	cohort1	date	Start date of cohort

Field	Source Table	Type	Description
end_date	cohort1	date	End date of cohort
email	cognito2	text	Learner email address
gender	cognito2	text	Gender of learner
UserCreateDate	cognito2	timestamp	Cognito profile creation date
birthdate	cognito2	text	DOB of learner
city, zip, state	cognito2	text	Location details
opportunity_name	opportunity1	text	Name of opportunity
opportunity_code	opportunity1	text	Code for opportunity

⏳ Step 2: Data Extraction

We used filtered SELECT and LEFT JOINs in PostgreSQL to combine relevant columns:

```sql
SELECT ...
FROM learneropp1 lo
LEFT JOIN learner1 l ON lo.enrollment_id = l.learner_id
LEFT JOIN cohort1 co ON lo.assigned_cohort = co.cohort_code
LEFT JOIN cognito2 c ON l.learner_id = c.user_id
LEFT JOIN opportunity1 o ON lo.learner_id = o.opportunity_id;
```

🪮 Step 3: Data Transformation

Transformation	Description
Remove duplicates	Used `DISTINCT` and filtering logic
Fix nulls	Filtered out rows with NULL `apply_date`, `email`, etc.
Format timestamps	Converted epoch and ISO strings to readable format
Resolve joins	Corrected incorrect joins with `user_id` and `opportunity_id`
Flag invalid data	Retained orphan cohorts but flagged

🗜️Step 4: Load to Clean Table

```
CREATE TABLE master_table_clean AS
SELECT ...
FROM master_table_raw
WHERE apply_date IS NOT NULL
AND assigned_cohort IN (SELECT cohort_code FROM cohort1);
```

🔮Page 3: Validation & Refinement

🔗Step 1: Data Quality Checks

We ran the following validations:

- **Record Count Check**: 113,602 (raw) vs 57,966 (clean) – reduction due to cleaning.
- **Duplicate Check**:

```
SELECT enrollment_id, COUNT(*) FROM master_table_clean GROUP BY enrollment_id
HAVING COUNT(*) > 1;
```

- **Null Value Check**:

```
SELECT COUNT(*) FROM master_table_clean WHERE email IS NULL;
```

- **Date Validity Check**:

```
SELECT * FROM master_table_clean WHERE apply_date IS NULL;
```

- **Foreign Key Check**:

```
SELECT * FROM master_table_clean WHERE assigned_cohort NOT IN (SELECT
cohort_code FROM cohort1);
```

🕐Step 2: Refinement Summary

Issue	Resolution
Null emails	Join fixed using correct `user_id` key
Invalid `apply_date`	Removed 3 rows with malformed dates

Issue	Resolution
Orphan `assigned_cohort`	Left in, but documented as invalid

📈Final Status Summary

Check	Status
Duplicate Records	Passed
Missing Values	Resolved
Date Format	Validated
Foreign Key Relationships	Flagged but documented

The dataset is now **clean, validated, and ready** for Week 3 visualization tasks.